

Plateau Mechanism, Warm-Start Linearity, and Consistency

Executive findings

The quantity that directly controls the normalized mean-mode curve is

$$q_N = R(a_N) - a_N, \quad a_N = \frac{h_2 \mu_{2,N} - h_1 \mu_{1,N}}{h_1 h_2}, \quad M_N = \sqrt{\pi} h_N |q_N|.$$

Here $\mu_{i,N}$ are the layer means before Appendix B, R is the code's nearest-integer map, and M_N is the exact contribution of the post-normalization constant relative-mean error to the discrete H_h^2 norm. The deepest symmetry-invariant phase variable is the class $[a_N] \in \mathbb{R}/\mathbb{Z}$; q_N is its centered representative. The raw mean μ_N is not invariant under an allowed integer translation and is therefore a driver and diagnostic, not the deepest state variable.

The exact plateau criterion is

$$\frac{M_{2N}}{M_N} = \frac{2N+1}{4N+1} \frac{|q_{2N}|}{|q_N|} \approx 1.$$

Thus a plateau occurs when $|q_{2N}|$ grows by approximately the reciprocal mesh ratio, nearly a factor of two. Seed 3 does this from $N = 80$ through $N = 320$. Its raw endpoint mean is nearly independent of N , so its unwrapped index $a_N = -2\mu_N/h_N$ grows approximately like $1/h_N$. Nearest-integer centering folds this growth into a wrapped-doubling sawtooth in q_N . This is the mechanism behind the plateau.

The continuum-sampled track has a different hierarchy. It starts with zero layer means, its raw endpoint mean is only $O(h^3)$ empirically, its $q_N = 2\bar{u}_{1,N}/h_N$ is $O(h^2)$, and its mean mode is negligible. The mean mode contributes 0.612% of the total norm at $N = 20$ and only 0.000631% at $N = 640$. Its error is therefore controlled by the nonconstant profile error, whose fitted slope is -0.9969 .

The plotted H_h^2 quantity is a **solution error**, not a consistency residual. The actual manuscript residual is

$$r_N = \Delta_h u_N^c - F_N(u_N^c),$$

and belongs naturally to ℓ_h^2 or ℓ^∞ , not H_h^2 . A direct calculation gives slopes -0.9956 in ℓ_h^2 and -0.9982 in ℓ^∞ , with $\|r_N\|_{\ell_h^2}/h_N \approx 0.0380$. This residual is common to both solution tracks because it depends only on the sampled continuum state. The warm error follows it with an almost constant ratio near 1.725, which is strong numerical evidence of a stable first-order response, but not a proof of the manuscript's uniform stability-gap hypothesis.

1. Quantities and causal hierarchy

1.1 Optimization and normalization

The live atomistic solver optimizes $2N$ reduced coordinates and reconstructs a full state of length $2N+1$ at every objective evaluation. This enforces

$$\bar{u}_1 + \bar{u}_2 = 0,$$

while leaving the relative mean $(\mu, -\mu)$ free. Appendix B is applied once after the optimizer returns. It chooses the integer $d_N = R(a_N)$ and a common translation so that the normalized means satisfy

$$m_N := \bar{u}_{1,N}^{\text{post}} = -\bar{u}_{2,N}^{\text{post}} = \frac{h_N}{2} q_N, \text{quad} |q_N| \leq \frac{1}{2}.$$

These are exact consequences of the manuscript’s discrete translation symmetry and Appendix B construction.¹ The live implementation uses `numpy rint`, which resolves half-integer ties by choosing the even integer.²

1.2 Why the quotient phase is deeper than the raw mean

In the sum-of-means gauge, an allowed integer translation by k changes the raw first-layer mean and phase index by

$$\mu'_N = \mu_N + \frac{h_N}{2}k, \quad a'_N = -\frac{2\mu'_N}{h_N} = a_N - k.$$

Therefore $[a'_N] = [a_N]$ modulo $\mathbb{Z}$. For a translation-equivariant nearest-integer rule,

$$q'_N = R(a_N - k) - (a_N - k) = q_N.$$

This proves the following hierarchy:

$$\boxed{\text{local stationary symmetry orbit} \longrightarrow [a_N] \in \mathbb{R}/\mathbb{Z} \longrightarrow q_N \longrightarrow m_N \longrightarrow M_N.}$$

The pre-solve Appendix B A/B test confirms this structure numerically. It changed the raw representative entering L-BFGS-B, but the pre- and post-solve integer shifts composed to the same total production shift and left the final q_N , M_N , energy, and continuum error nearly unchanged.³

1.3 Exact condition for a plateau

For $h_N = 4\pi/(2N + 1)$,

$$M_N = \sqrt{\pi}h_N|q_N|.$$

Consequently,

$$\boxed{M_{2N} \approx M_N \iff \frac{|q_{2N}|}{|q_N|} \approx \frac{4N + 1}{2N + 1} \approx 2.}$$

The plateau is not caused merely by q_N being nonzero. If q_N were approximately constant, M_N would approximately halve. A plateau needs $|q_N|$ to grow at nearly the inverse mesh ratio before the centered remainder wraps.

2. Seed-3 mechanism

For seed 3, the pre-Appendix-B endpoint mean remains approximately 0.0433 to 0.0435 across all six grids. In the sum gauge,

$$a_N = -\frac{2\mu_N}{h_N} = -\frac{(2N + 1)\mu_N}{2\pi}.$$

The nearly N -independent nonzero μ_N therefore makes a_N approximately proportional to N . If

$$\varepsilon_N = a_{2N} - 2a_N,$$

the exact ties-to-even recurrence is

$$q_{2N} = 2q_N - \varepsilon_N - R(2q_N - \varepsilon_N).$$

The defect ε_N is small for the observed seed-3 branch, so the remainder follows approximate centered doubling.

¹Source 1 supplies the exact discrete symmetry, Appendix B mean bound, consistency residual, stability estimate, and convergence theorem.

²Source 2 supplies the live coordinate maps, objective gradient, error norm, solver settings, and all numerical arrays used here.

³Source 3 reports the six-grid pre-normalization A/B test; all endpoints passed the current production acceptance checks.

N	q_N	M_N	total H_h^2	mean share of norm
20	+0.282933	1.53704e-1	1.59089e-1	96.62%
40	-0.439450	1.20840e-1	1.24476e-1	97.08%
80	+0.113199	1.56604e-2	1.68737e-2	92.81%
160	+0.216269	1.50063e-2	1.56189e-2	96.08%
320	+0.422538	1.46823e-2	1.51285e-2	97.05%
640	-0.165304	2.87420e-3	3.02038e-3	95.16%

From 80 to 160 and from 160 to 320, the local mean-mode slopes are only -0.0616 and -0.0315 . At 640, the centered remainder wraps and the mean-mode slope becomes -2.3528 . The plot is therefore a genuine finite-grid sawtooth under an $O(h)$ envelope, not a universal period-four orbit.

The near constancy of the raw seed-3 mean is an important empirical driver, but it is not an L-BFGS-B invariant. A callback audit found a change of roughly 6.6×10^{-4} to 7.9×10^{-4} during optimization. Other random seeds changed materially more, so no general mean-conservation law is present.⁴

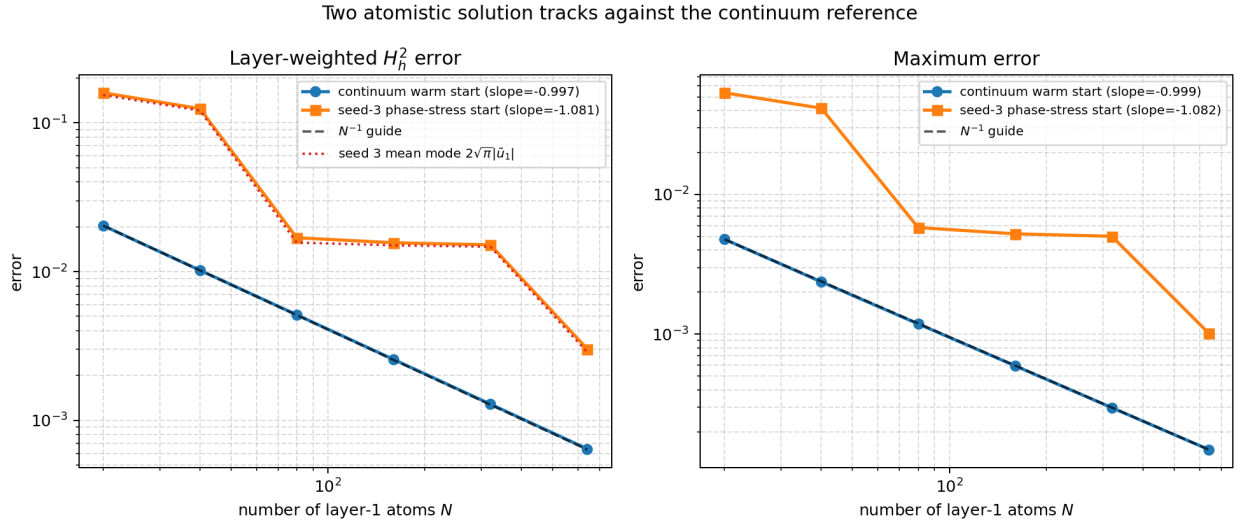


Figure 1: Production convergence plot. The red dotted seed-3 mean mode nearly overlays the seed-3 H_2 error.

3. Why the continuum-sampled track has no plateau

3.1 Relative mean before and after optimization

The sampled continuum pair is odd and its two discrete means vanish to machine precision. L-BFGS-B does not preserve these individual means exactly, but the drift is extremely small and decreases rapidly with refinement.

N	input $\bar{u}_1$	raw endpoint $\bar{u}_1$	q_N
20	+1.39e-18	+3.5170e-5	2.29497e-4
40	-2.52e-18	+4.7318e-6	6.10001e-5
80	-2.30e-18	+6.1398e-7	1.57326e-5
160	-5.55e-18	+7.8171e-8	3.99367e-6
320	-5.55e-18	+9.1204e-9	9.30450e-7
640	0	+1.1447e-9	2.33375e-7

The endpoint mean has fitted slope -2.9876 , approximately $O(h^3)$. Because $q_N = 2m_N/h_N$, the warm remainder is approximately $O(h^2)$ and decreases by about a factor of four under each doubling. This is the opposite of the seed-3 plateau condition.

⁴Source 4 reports accepted-iterate callback traces for the continuum start and seeds 0-4 across all six grids.

This $O(h^3)$ behavior is an empirical property of the current warm branch and reduced-coordinate solve, not a theorem established by the manuscript. The safe conclusion is that the optimizer does not preserve zero exactly, but the induced mean is too small to affect the headline convergence curve.

3.2 Mean contribution versus total error

The error decomposition is orthogonal because the constant component has zero first and second differences:

$$\|e_N\|_{H_h^2}^2 = M_N^2 + \|e_N - \bar{e}_N\|_{H_h^2}^2.$$

N	warm M_N	warm H_h^2	mean share	mean-free H_h^2
20	1.24675e-4	2.03682e-2	0.6121%	2.03679e-2
40	1.67737e-5	1.02044e-2	0.1644%	1.02044e-2
80	2.17650e-6	5.12025e-3	0.0425%	5.12025e-3
160	2.77110e-7	2.56641e-3	0.0108%	2.56641e-3
320	3.23310e-8	1.28517e-3	0.0025%	1.28517e-3
640	4.05779e-9	6.42871e-4	0.0006%	6.42871e-4

The warm mean mode does not dominate at any grid. The total and mean-free fitted slopes are -0.996919 and -0.996915 , respectively. Moreover, the separate L_h^2 , first-difference, and second-difference components have slopes -0.9976 , -0.9968 , and -0.9963 . The nearly straight curve is therefore not created by cancellation between irregular components; all three nonconstant components are individually first order.

4. Solution error versus consistency residual

4.1 Correct definitions

The manuscript defines the sampled-continuum consistency residual by

$$r_N^i = \Delta_i u_N^{c,i} - F_N^i(u_N^c).$$

Theorem 4 proves an $O(h)$ bound in ℓ_h^2 and ℓ^∞ . Theorem 3 then bounds a discrete solution error by the residual plus layer-mean terms under a uniform stability condition. Theorem 5 combines those statements to obtain an $O(h)$ discrete H_h^2 solution error.⁵

Therefore the phrase “ H^2 error for the consistency residual” mixes two different objects:

- $\|u_N^d - u_N^c\|_{H_h^2}$ is the plotted solution error;
- $\|\Delta_h u_N^c - F_N(u_N^c)\|_{\ell_h^2}$ is the consistency residual.

The residual is not attached to an optimizer start. There is one sampled continuum state at each N , so the same r_N is relevant to the warm and seed-3 comparisons.

4.2 Direct residual calculation

The residual was evaluated by inserting the saved 800-point continuum spline, sampled on each native lattice, into the production atomistic operator. The calculation used the paper identity

$$r_N^i = -\frac{1}{h_i} \frac{\partial E_N}{\partial u_i}(u_N^c).$$

An independent implementation of $\Delta_i u_N^{c,i} - F_N^i(u_N^c)$ agreed with this gradient formula to at most 1.17×10^{-13} .

⁵Source 1 supplies the exact discrete symmetry, Appendix B mean bound, consistency residual, stability estimate, and convergence theorem.

N	$\ r_N\ _{\ell_h^2}$	residual / h_N	$\ r_N\ _\infty$
20	1.173893e-2	0.0383003	5.078409e-3
40	5.905159e-3	0.0380633	2.527345e-3
80	2.966139e-3	0.0380021	1.258925e-3
160	1.487086e-3	0.0379867	6.291833e-4
320	7.446404e-4	0.0379835	3.153028e-4
640	3.726424e-4	0.0379867	1.603142e-4

The all-grid slopes are

$$\|r_N\|_{\ell_h^2} : -0.995572, \quad \|r_N\|_\infty : -0.998174.$$

The adjacent ℓ_h^2 slopes range only from -0.9913 to -0.9988 . The actual consistency residual is therefore nearly perfectly first order on these grids.

This result is insensitive to the production approximations. Replacing the image cutoff $M = 80$ by $M = 160$ changes the residual by at most 3.68×10^{-13} in ℓ_h^2 . Replacing the 800-point continuum reference by the 400-point reference changes the residual norm by at most 2.69×10^{-7} .⁶

4.3 What the linear residual explains

For the continuum-sampled track,

$$\frac{\|u_N^{\text{warm}} - u_N^c\|_{H_h^2}}{\|r_N\|_{\ell_h^2}} = 1.7351, 1.7280, 1.7262, 1.7258, 1.7259, 1.7252.$$

The nearly constant ratio, negligible mean term, and matching slopes strongly support the interpretation that the warm endpoint follows one stable local branch responding linearly to the first-order consistency defect. This is exactly the structure suggested by the manuscript’s stability estimate. However, it does not verify the uniform gap $1 - C_P L_F > 0$, prove global minimality, or replace the distinction between a computed stationary endpoint and the exact u_N^d in the theorem.

For seed 3, the consistency residual remains the same nearly linear sequence, but the solution error does not inherit a straight line. Its centered mean remainder adds the dominant sawtooth term. Even after removing that mean, the seed-3 mean-free solution error has an overall slope -1.0563 , but its adjacent slopes are

$$-0.459, -2.249, -0.537, -0.248, -1.974.$$

It is broadly first order over all six grids but is not locally proportional to the consistency residual. This remaining irregularity is evidence that the seed-3 endpoints contain a branch- or phase-dependent profile difference in addition to the constant mean mode. It must not be relabeled as a consistency residual.

5. Comparative summary

quantity	continuum-sampled	seed 3	interpretation
raw endpoint mean	approximately $O(h^3)$	approximately nonzero constant	empirical branch behavior
centered q_N	approximately $O(h^2)$	wrapped-doubling sawtooth	deepest canonical phase diagnostic
mean mode M_N	slope -2.9876	sawtooth; plateau at 80-320	exact $\sqrt{\pi}h q $ contribution
mean share of H_h^2	0.612% down to 0.0006%	92.8% to 97.1%	negligible versus dominant

⁶Source 2 supplies the live coordinate maps, objective gradient, error norm, solver settings, and all numerical arrays used here.

quantity	continuum-sampled	seed 3	interpretation
mean-free solution error	slope -0.9969 , locally straight	slope -1.0563 , locally irregular	profile-shape difference
full solution error	slope -0.9969 , locally straight	slope -1.0808 , sawtooth	plotted curves
consistency residual	one common sequence, slope -0.9956	same sequence	property of u_N^c , not the start

The causal distinction is now sharp:

1. **Consistency** supplies a common, nearly linear $O(h)$ forcing error.
2. **Local stability and branch selection** determine how an atomistic endpoint responds to that forcing.
3. **Quotient phase** $[a_N]$, represented by q_N , determines the normalized constant relative-mean component.
4. **Seed 3** has a nonzero raw phase branch whose centered remainder produces the plateau.
5. **The warm track** stays near the odd zero-mean branch, so its mean mode is negligible and the common consistency scaling is visible directly in the solution error.

6. Recommendation and limitations

The most informative follow-up diagnostics are q_N , M_N , the mean-free solution error, and the actual ℓ_h^2 consistency residual. They should be plotted separately and named distinctly. The production workflow should continue to enforce the smooth sum-of-means gauge during optimization and apply Appendix B after convergence.

The current evidence does not support forcing both layer means to zero or canonicalizing with a discontinuous integer roll inside L-BFGS-B. SciPy’s interface updates the supplied parameter vector and uses limited-memory variable-metric corrections; no relative-mean constraint is present in the live call.⁷ The original method is a limited-memory quasi-Newton algorithm for componentwise bounds, not an optimizer on the quotient by the Appendix B symmetry.⁸

The computed residual uses the production finite image cutoff and the saved numerical continuum reference. The cutoff and 400/800 checks show these errors are negligible at the reported scale. The six-grid experiments remain finite-resolution evidence. They do not prove a universal sawtooth schedule, seed-3 branch membership, global minimality, or the manuscript’s uniform stability gap.

Sources

1. *Continuum Limit of a One-Dimensional Atomistic Model for Moire Relaxation*, local manuscript Hao_OralExam_Continuum_Limit.pdf, July 21, 2026. Equations (15), (23), (26), (38)-(39), (53), and Appendix B.
2. Live CLR implementation and saved data, accessed September 8, 2026. Main script: `atomistic_two_chain_lj_lbfgs_convergence_study.py`. Production arrays and table: `profiles.npz` and `convergence_summary.csv`.
3. *One-Time Pre-Solve Appendix B Normalization: Seed-3 A/B Test*, external read-only experiment report, September 8, 2026.
4. *Does L-BFGS-B Preserve the Relative Mean?*, local deep-research report `lbfgsb_relative_mean_preservation_deep_research_report.pdf`, September 8, 2026.
5. SciPy. `scipy.optimize.fmin_l_bfgs_b`, current reference documentation, accessed September 8, 2026.
6. Richard H. Byrd, Peihuang Lu, Jorge Nocedal, and Ciyu Zhu. “A Limited Memory Algorithm for Bound Constrained Optimization.” *SIAM Journal on Scientific Computing* 16, no. 5 (1995): 1190-1208.

⁷Source 5 documents the callback, stopping rules, componentwise bounds, and limited-memory parameter of SciPy’s interface.

⁸Source 6 is the primary algorithm paper for the limited-memory BFGS method used by L-BFGS-B.